



Spaceflight affects your eyes

70 percent of astronauts who spend time on the ISS experience swelling at the back of their eyes. <https://digit.in/feb22-10>



Google urges Apple

Google executive urges Apple to incorporate RCS into iMessage. <https://digit.in/feb22-13>

collecting “more data than was needed” from users. It denied its involvement in the scandal, but agreed to pay \$58 million to any customers who had a bank account linked to any of its 5,000 client apps. It was also reported that the corporation got users’ bank login information through its own “Plaid Link” interface, which mimicked the user’s own bank account login screen. Irrespective of the scale of the scam, it is a Not in our books.



LOW-CARBON SHIPPING

Ever wonder how your Amazon Prime package gets delivered to you within a day?

People assume it is a feat of miraculous logistics (which it is) and leave it at that. But what about the environmental cost? No one can plant enough trees to offset the carbon impact of a singular one-day delivery, much less the daily packages that deliver as a result of a shopping spree. Well, to remedy that, transporters are pledging to reduce greenhouse gas emissions to zero by the middle of the century, but only if governments put an obligatory fee on shipping fuel to support the development of new low-carbon technology. The plans were filed to the International Maritime Organisation (IMO), a global regulator and part of the United Nations, by the International Chamber of Shipping (ICS), which represents the bulk of the global shipping industry. The shipping industry association proposed a carbon tax, a new fund for the development of low-carbon technology, and a 25 percent reduction in emissions by 2030, with zero emissions by 2050. A definite Hot!



RENEWABLE FUEL

A research team born at ETH Zurich has analysed various fuel options for cargo

ships operating within Europe. The researchers’ plan was to use real-world shipping data from 2018 to determine the impact of replacing each ship’s propulsion system. The researchers examined hydrogen, ammonia, methane, methanol, and diesel propul-

sion systems, all of which are made from renewable feedstock and run on clean electricity. This includes using CO₂ extracted from the atmosphere to make methane, methanol, or fuel. However, most of these techniques or the production of these rely on immature technologies at some point in their production cycle. They are also all speculative to varying degrees. When it comes to their price tags, there is a lot of guesswork. For instance, atmospheric carbon capture is still in its early stages, with only a few demonstration plants functioning around the world, making it prohibitively expensive. So, while research scientists are headed in the right direction and renewable fuel is somewhat implemented already, there is still a long way to go. The effort and progress is definitely Hot, though.



SELF-FERTILISING CROPS

Every year, roughly 110 million tonnes of nitrogen fertiliser are utilised in global crop production. Plants such as soy and beans, which are members of the legume family, which also includes peas and lentils, employ a “clever approach” to make their own nitrogen, according to the Top 10 Emerging Technologies of 2021 report published by Scientific American and the World Economic Forum. Other crops, such as corn and other grains, may now be “coaxed” to self-fertilise, according to scientists. There is some level of natural self-fertilisation that takes place in legumes, due to the help of a microbe also known as “nitrogen-fixer”. To emulate something similar in other crops such as rice, scientists are aiming to create synthetically engineered nitrogen-fixing bacteria that can survive on any plant and make their own fertiliser, effectively eliminating the need for fossil fuel-based artificial fertilisers. Definitely Hot!



BROWSE AD-FREE

With Google taking multiple measures to ban targeted ads for minors, other tech giants might be forced to follow suit by law. The European Parliament has

approved a bill that looks to ban some targeted ads. The proposal also banned dark patterns, which is used by some websites to deceive users into agreeing to disclose their data. Companies who violate these policies could face fines of up to 6 percent of their global revenue. More obstacles remain in the way of the Digital Services Act; negotiations with the European Council began on January 31st 2022. A similar law was recently presented in the United States by Democrats, which, if passed, would similarly prohibit targeted advertising. While such legal protective measures are from being a reality in India, one step forward somewhere else will create a ripple effect which will eventually end up in a similar bill passed in India. Hot, again!



VEGAN SUNNY SIDE UP

With veganism seeing a rise, scientists and researchers are more motivated to work

on nutritious vegan alternatives for common protein sources. Eggs get a lot of attention since they are so versatile. While there is no singular multi-purpose egg alternative, vegans have managed to collect different types of eggs such as hard-boiled, poached, scrambled etc over the years in the form of different egg substitutes. Something that was lacking so far in the vegan scene was a sunny-side up, because it is very difficult to emulate the runny yolk that egg-lovers desire. However, through trial and error, there are a couple decent substitutes for a sunny-side-up. The most popular one includes two internet-famous items: Vegan Egg, an egg white substitute and pumpkin puree, the famous festive ingredient. Vegan Egg powder is mixed with warm water and laid around in a pan with a hole in the middle for the yolk. After some time, a spoonful of pumpkin puree is gently dropped into the centre. The exterior of the puree blob cooks like a flatbread with a flaky-ish crust, but the interior is still soft and gooey. It makes for the perfect sunny side up. Vegan alternatives encourage sustainability, and we find that very Hot. **d**